

CO 250 Review 7: Geometry of Simplex

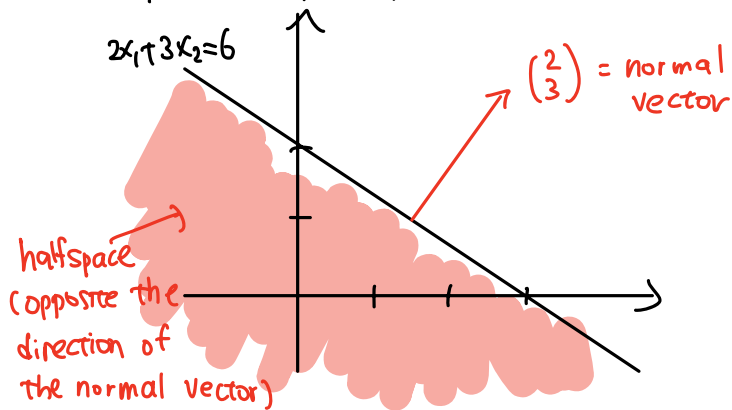
Geometry of LPs and Simplex : The feasible region of an LP is a polyhedron, and Simplex goes from one extreme point to another in the polyhedron.

① The feasible region of an LP is a polyhedron

A polyhedron is a set of the form $\{x \in \mathbb{R}^n : Ax \leq b\}$.

Each inequality $a_i^T x \leq b_i$ is a halfspace.

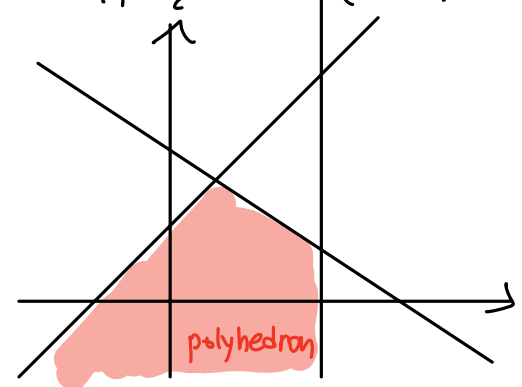
Example: $2x_1 + 3x_2 \leq 6$



Example:

$$\begin{aligned} 2x_1 + 3x_2 &\leq 6 \\ x_1 &\leq 2 \\ -x_1 + x_2 &\leq 1 \end{aligned}$$

$$\begin{pmatrix} 2 & 3 \\ 1 & 0 \\ -1 & 1 \end{pmatrix} x \leq \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}$$



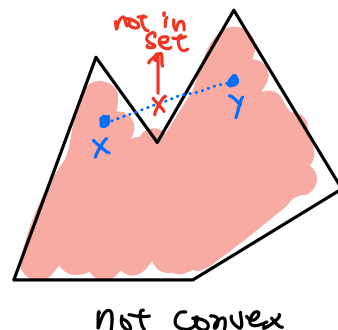
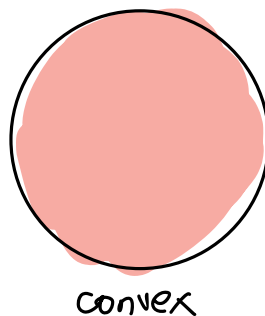
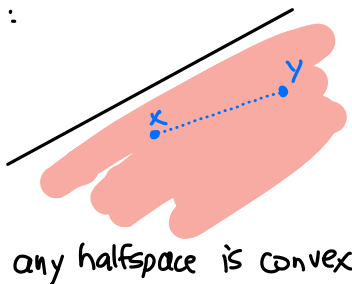
A polyhedron is the intersection of halfspaces. Any constraint can be turned into a halfspace $[a^T x \geq b \Rightarrow -a^T x \leq -b \quad a^T x = b \Rightarrow a^T x \leq b, -a^T x \leq -b]$

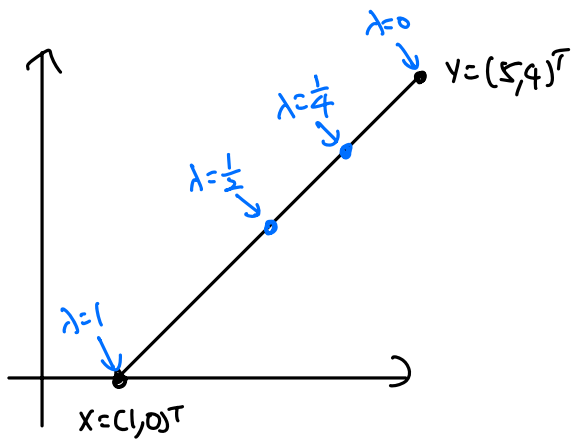
② A polyhedron is convex

A set S is convex if for any $x, y \in S$, the line segment from x to y is entirely in S .

More formally: if $x, y \in S$, then $\lambda x + (1-\lambda)y \in S$ for all $\lambda \in [0, 1]$.

Examples:





$$\lambda(1, 0)^T + (1-\lambda)(5, 4)^T$$

$$\lambda=0: (5, 4)^T$$

$$\lambda=1: (1, 0)^T$$

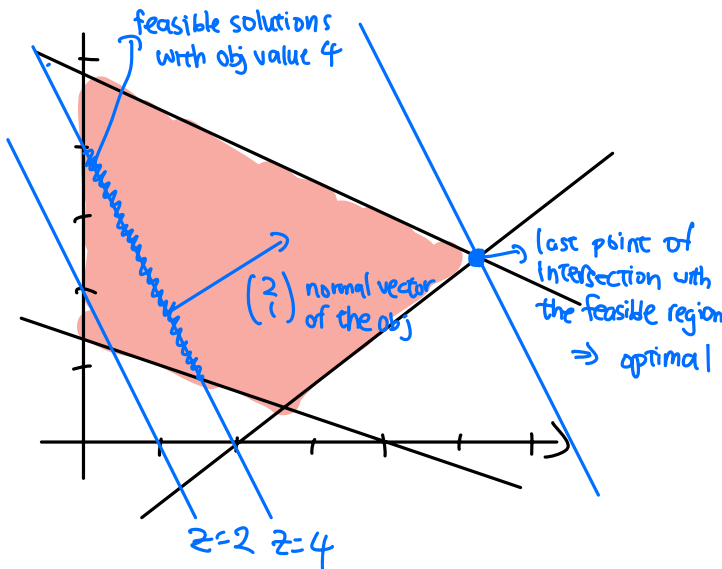
$$\lambda=\frac{1}{2}: (3, 2)^T$$

$$\lambda=\frac{1}{4}: (4, 3)^T$$

Facts: The intersection of convex sets is convex. So a polyhedron is convex. Hence the feasible region of any LP is convex.

③ Geometry of the objective function (optimal solutions and extreme points)

Consider an LP in $\mathbb{R}^2$ with this feasible region:



Suppose the objective function is

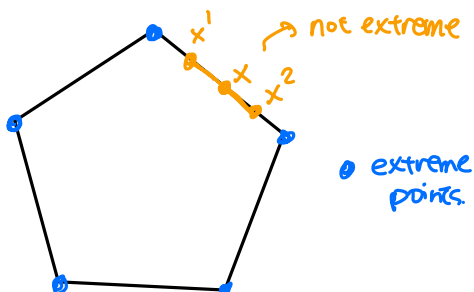
$$\max 2x_1 + x_2$$

$$\text{Let } z = 2x_1 + x_2$$

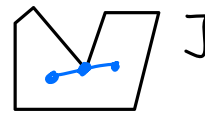
Pull the line towards $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ as far as possible.

This optimal solution is an "extreme point" of the polyhedron. (In general, there can be optimal solutions that are not extreme points.)

Given a convex set C and $x \in C$, x is NOT an extreme point of C if x is strictly inside a line segment of two other points in C .



[Aside: Extreme points are defined for convex sets only.]



④ Extreme points of a polyhedron (an algebraic characterization)

In $\mathbb{R}^2$, extreme points seem to be the intersection of ≥ 2 non-parallel lines.

Generalizing, in $\mathbb{R}^n$, an extreme point is the intersection of $2n$ "independent" hyperplanes.

A point in $\mathbb{R}^n$ is on a line if it satisfies the constraint $a_i^T x \leq b_i$ with equality $a_i^T x = b_i$.

For a constraint $a^T x \leq b$ and a point $\bar{x}$, it is a tight constraint if $a^T \bar{x} = b$.

Example: $\begin{pmatrix} 1 & 0 & 2 \\ -2 & 1 & 0 \\ 0 & 2 & -1 \end{pmatrix} x \leq \begin{pmatrix} 7 \\ 2 \\ 1 \end{pmatrix}$ For $\bar{x} = (1, 2, 3)^T$, $\begin{pmatrix} 1 & 0 & 2 \\ -2 & 1 & 0 \\ 0 & 2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 0 \\ 1 \end{pmatrix} \leq \begin{pmatrix} 7 \\ 2 \\ 1 \end{pmatrix}$ → tight
→ tight

Theorem: Let $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a polyhedron, and let $\bar{x} \in P$. Suppose $A^T \bar{x} \leq b^T$ are all tight constraints for $\bar{x}$. Then $\bar{x}$ is an extreme point of P if and only if $\text{rank}(A^T) = n$.

Example: $\begin{pmatrix} 3 & 2 \\ 1 & 1 \\ 2 & 2 \end{pmatrix} x \leq \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}$ For $\bar{x} = (1, -1)^T$, the tight constraints are $\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} x \leq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$
 $\text{Rank} \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} = 1$, so $\bar{x}$ is not an extreme point.

For $\bar{x} = (3, -3)^T$, all constraints are tight. So $A^T = \begin{pmatrix} 3 & 2 \\ 1 & 1 \\ 2 & 2 \end{pmatrix}$.

Since $\text{rank}(A^T) = 2$, $\bar{x}$ is an extreme point.

⑤ Extreme points for LPs in SEF: extreme points $\Leftrightarrow$ BFS.

Theorem: Let $P = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}$ (this is in SEF). Let $\bar{x} \in P$. Then $\bar{x}$ is an extreme point of P if and only if $\bar{x}$ is a basic feasible solution.

Example: $\max (1 \ 2 \ 3) x$
s.t. $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{pmatrix} x = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$ $\{1, 2\}$ is a basis with BFS $(5, 0, 0)^T$.
 $x \geq 0$

Why is this an extreme point? First convert constraints into $\square x \leq \square$.

For equality constraints: $\square x = \square \Rightarrow \begin{matrix} \square x \leq \square \\ -\square x \leq -\square \end{matrix}$, $x \geq 0 \Rightarrow -x \leq 0$.

$(1 \ 0 \ 1)x = 5$
 $(0 \ 1 \ 2)x = 10$
 $x \geq 0$

$$\begin{pmatrix}
 1 & 0 & 1 \\
 -1 & 0 & -1 \\
 0 & 1 & 2 \\
 0 & -1 & -2 \\
 -1 & 0 & 0 \\
 0 & -1 & 0 \\
 0 & 0 & -1
 \end{pmatrix} x \leq \begin{pmatrix} 5 \\ -5 \\ 10 \\ -10 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

tight constraints

$$A = \begin{pmatrix}
 1 & 0 & 1 \\
 -1 & 0 & -1 \\
 0 & 1 & 2 \\
 0 & -1 & -2 \\
 0 & 0 & -1
 \end{pmatrix} \quad \text{rank } 3.$$

$\Rightarrow (5, 10, 0)^T$ is an extreme point.

⑥ Relation to Simplex

Recall: Simplex moves from one BFS to another BFS. Geometrically, this means moving from one extreme point to another extreme point, in the direction of the objective.